The quartic analogue over $\mathbb{Z}[i]$ of the cubic-theta zero-free half-plane argument, and where it stops

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Abstract

In the OpenAI zero-free half-plane argument the boundary 11/12 of the first stage is $1-\frac{1}{2}\cdot\frac{1}{6}$, the $\frac{1}{6}$ being the order of the sextic residue symbol used to embed the Möbius sum into a family in which the Gauss–Jacobi relations convert the Möbius function into cubic Gauss sums, the Fourier coefficients of Kubota's cubic theta function. We show that the arithmetic half of this mechanism goes through verbatim with the quartic symbol over $\mathbb{Z}[i]$: for squarefree primary n, $\mu(n)\tilde{g}_{-1}(n)=\chi_n(-1)\overline{\alpha(n)}\tilde{g}_1(n)$, so the quartic family would start at $1-\frac{1}{2}\cdot\frac{1}{4}=7/8$ before any refinement. No automorphic input of the required shape is known: the quartic Gauss sums are Whittaker coefficients of the distinguished theta function on the 4-fold cover of $\mathrm{GL}_3(\mathbb{Q}(i))$, not of a form on a cover of GL_2, and the twisted Dirichlet series they generate has conductor $\mathrm{N}(h)^3$ in the row variable, so its functional equation lengthens the column sum instead of shortening it. We state the missing input in one sentence; we do not prove that it cannot exist.

1 The sextic mechanism

We follow the 11/12 manuscript [13, §2, §4.1, App. A.1] (labels: [EST] checked in the source, [DERIVED] computed here, [LIT] quoted, [JUDG] judgement). Over $O=\mathbb{Z}[\omega]$ the Möbius sum $A_1(D)=\sum\mu(n)\nu(n)W(\mathrm{N}n/D)$ is embedded in $A_u(D)=\sum\mu(n)\nu(n)\chi_n(u)W(\mathrm{N}n/D)$, $\chi_n=(\cdot/n)_6$. Three facts carry the argument [EST]:

1. *Amplifier.* $\chi_n(p^6)=\mathbf{1}_{p\nmid n}$, so the rows $u=p^6$, $Y/2<\mathrm{N}p\le Y$, reproduce A_1 up to $O(D/Y)$; with $Y=H^{1/6}$ and the mean square $\sum_{\mathrm{N}u\le H}|A_u|^2\ll D^{1+\varepsilon}H$ this gives $A_1\ll D^{11/12+\varepsilon}$ (eqs. (2.4)–(2.5) there).

2. *Möbius absorption.* With normalised Gauss sums $\gamma_j(n)=\mathrm{N}(n)^{-1/2}\sum_x\chi_n(x)^j e(x/n)$ and $\alpha(n)=n/|n|$, Lemma 4.1 there gives $\gamma_2(n)^3=\mu(n)\alpha(n)$ and $\mu(n)\gamma_{-1}(n)=\chi_n(-1)G(n)^{-1}\alpha(n)\gamma_2(n)$, G an explicit ray-class factor; the sign -1 per prime comes from $J(\chi^2,\chi^2)=-p$.

3. *Automorphy.* $\gamma_2(n)$, a cubic Gauss sum, is the Fourier coefficient at $\lambda^{-3}n$ of Kubota's cubic theta function on the 3-fold cover of $\mathrm{GL}_2(\mathbb{Q}(\omega))$ (Patterson [14]); the theta transformation of Dunn–Radziwiłł [4] sends column scale X to dual scale $\mathrm{N}(h)^2/X$ with the dual twist $\chi_h^{-1}\chi_h^{-2}=\chi_h^3$ quadratic (eq. (2.14) there), so the quadratic large sieve of Goldmakher–Louvel applies with the factor $M+L$.

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2 The arithmetic half goes through over $\mathbb{Z}[i]$

Let $O = \mathbb{Z}[i]$; call n primary if $n \equiv 1 \pmod{2+2i}$ (unique primary generator per odd ideal). For a primary prime π of norm $p \equiv 1$ (4) let $\chi_\pi(x) \equiv x^{(p-1)/4} \pmod{\pi} \in \mu_4 \subset O$, extended multiplicatively in the denominator; $e(z) = \exp(2\pi i \operatorname{Tr} z)$; $\tilde{g}_j(n) = \mathrm{N}(n)^{-1/2} \sum_{x \bmod n} \chi_n(x)^j e(x/n)$ for $j = 1, 2, 3$ ($\tilde{g}_3 = \tilde{g}_{-1}$); $\alpha(n) = n/|n|$.

Lemma 1 (quartic Möbius absorption)**.** *For squarefree primary $n \in \mathbb{Z}[i]$:*

(i) $\tilde{g}_1(n)^2 = \mu(n)\alpha(n)$;

(ii) $\tilde{g}_2(n) = \chi_n(-1) = (-1)^{(\mathrm{N}n-1)/4}$, *the ray-class character modulo* 4;

(iii) $\tilde{g}_1(n)\tilde{g}_{-1}(n) = \chi_n(-1)$;

(iv) $\mu(n)\,\tilde{g}_{-1}(n) = \chi_n(-1)\,\overline{\alpha(n)}\,\tilde{g}_1(n)$.

Proof. At a prime: $\tilde{g}_1^2 = (J(\chi,\chi)/\sqrt{p})\,\tilde{g}_2$ and $J(\chi_\pi,\chi_\pi) = -\chi_\pi(-1)\pi$ for primary π [9, Ch. 9, Prop. 9.9.4; equivalently Prop. 9.10.1] (cf. [2, (3.11)]); since $O/\pi \cong \mathbb{F}_p$ with $\operatorname{Tr}(x/\pi) = 2ax/p$ ($\pi = a + bi$), $\tilde{g}_2(\pi) = \sum_x (x/p)e(2ax/p)/\sqrt{p} = (2a/p) = (2/p) = (-1)^{(p-1)/4} = \chi_\pi(-1)$, using $(a/p) = (p/|a|) = (b^2/|a|) = 1$ (as $p \equiv 1$ (4), $p \equiv b^2$ (a)) and $(2/p) = (-1)^{(p-1)/4}$ for $p \equiv 1$ (4); this is (ii); hence $\tilde{g}_1^2 = -\chi(-1)\alpha\chi(-1) = -\alpha = \mu(\pi)\alpha(\pi)$, which is (i); (iii) is $g(\bar{\chi}) = \chi(-1)\overline{g(\chi)}$ together with $|\tilde{g}_1(\pi)| = 1$ (the additive character $e(\operatorname{Tr}(x/n))$ is primitive modulo odd n); (iv) is (iii) divided by (i). For an inert prime $\pi = -q$, $q \equiv 3$ (4), $O/\pi = \mathbb{F}_q(i)$ and $\chi_\pi(x) \equiv x^{(q^2-1)/4}$ is trivial on $\mathbb{F}_q^\times$ (as $(q+1)/2$ is even); grouping x by $\mathbb{F}_q^\times$-cosets, $\sum_{t \in \mathbb{F}_q^\times} e(\operatorname{Tr}(tc/\pi)) = q\mathbf{1}_{\operatorname{Tr} c = 0} - 1$, so $\tilde{g}_j(\pi) = \chi_\pi(i)^j$ for $j = 1, 3$ and $\tilde{g}_2(\pi) = 1$; here $\chi_\pi(i) = i^{(q^2-1)/4} = \pm 1$, $\chi_\pi(-1) = 1$ and $\mu(\pi)\alpha(\pi) = (-1)(-1) = 1$, so (i)–(iv) hold. For squarefree $n = ab$, $(a,b) = 1$, the Chinese remainder theorem gives $\tilde{g}_j(ab) = \tilde{g}_j(a)\tilde{g}_j(b)\big(\chi_a(b)\chi_b(a)\big)^j$ and quartic reciprocity for primary a, b gives $\chi_a(b) = \pm\chi_b(a)$, hence $\epsilon := \chi_a(b)\chi_b(a) = \pm\chi_b(a)^2 = \pm 1$. Then $\tilde{g}_j(ab) = \tilde{g}_j(a)\tilde{g}_j(b)\epsilon^j$ with $\epsilon^{-1} = \epsilon$, so both sides of (i)–(iii) are multiplicative and both sides of (iv) pick up the same factor ϵ; hence (i)–(iv) hold for squarefree primary n. □

[DERIVED] Identities (i)–(iv) were verified numerically for all split primary primes of norm ≤ 73 and for squarefree n with two and three split prime factors up to norm 18241, with $|\mathrm{LHS} - \mathrm{RHS}| \leq 5 \cdot 10^{-15}$ (`lab/quartic/mobius_absorption_check.py`; the prime-level relations $\tilde{g}_1^2 = J\tilde{g}_2$, $J = -\chi(-1)\pi$ independently in `gauss_identity_check.py`), and for the inert primes $-3, -7, -11$ and products of inert and split primes up to norm 585 with errors $\leq 3 \cdot 10^{-13}$ (`inert_check.py`).

So the Möbius function is absorbed exactly as in the sextic case, with the explicit factor $\chi_n(-1)$ in place of $\chi_n(-1)G(n)^{-1}$. The one difference: the surviving Gauss sum $\tilde{g}_1$ is still of order 4 (the cubic case halves $6 \to 3$; here the order-2 sum is a constant and carries nothing).

Remark 2 (amplifier arithmetic)**.** *[DERIVED] Rows $u = p^4$ reproduce A_1; there are $\asymp H^{1/4}/\log H$ of them. If $\sum_{\mathrm{N}u \leq H} |A_u(D)|^2 \ll D^{1+\varepsilon}H$ held, then $|A_1|^2 \ll D^{1+\varepsilon}H^{3/4} + D^2H^{-1/2}$, and with $H = D^{1+\vartheta}$, $A_1 \ll D^{7/8+3\vartheta/8+\varepsilon}$: the first stage would start at $7/8 = 1 - \frac{1}{2}\cdot\frac{1}{4}$ instead of $11/12$. (General pattern: a family of order $2k$ gives $1 - 1/(4k)$; $k = 1$ is the quadratic case, see Remark 3.) [JUDG] Replacing sixth by fourth powers in the bookkeeping of the $7/8$ paper would formally turn $\sigma_0 = 1 - h/6 + b/12$ into $1 - h/4 + b/8$, which at the first-stage values $(h, b) = (1/2, 0)$ returns $7/8$; but the two energy constraints that cap the slot length and the row counts of [11] are tied to the sixth-power structure and would have to be rederived, so this is not a prediction of a refined boundary.*

3 The automorphic half fails

Everything below concerns the column sum that the mean square produces after Poisson summation in u, which by Lemma 1 is, up to the explicit factors,

$$B_h(X) = \sum_{n \text{ sqfree primary}} \tilde{g}_1(n)\,\overline{\alpha(n)}\,\xi(n)\,\chi_n(h)\,U(\mathrm{N}n/X), \tag{1}$$

and Step 4 of [13] needs a transformation of (1) to a dual sum of length about $\mathrm{N}(h)^2/X$ with a quadratic twist in h.

(a) [LIT] Kazhdan–Patterson [10, Introduction and Thm. II.2.5]: the theta representations on the n-fold cover of GL_r "have a unique Whittaker model only when $r = n-1$ or n". For $n = 4$ the theta on the 4-fold cover of GL_2 is therefore not distinguished; the distinguished quartic theta lives on the 4-fold cover of GL_3 (for covers with odd c, [6, §2]).

(b) [LIT] The Fourier coefficients of the biquadratic theta on $\mathrm{GL}_2(\mathbb{Q}(i))$ are not quartic Gauss sums; at squarefree indices they are known only conjecturally (Patterson [15]; Eckhardt–Patterson [5, Conj. 2.11]), with Suzuki's results at squares, cubes and fourth powers [16]; David–Dunn–Hamieh–Lin [2, §1.2]: "in general not all determined! This is in stark contrast to the case of the cubic theta function". Even if Patterson's conjecture were known, the coefficient at a prime would be a square root of the Gauss sum, so this object could not play the role of Kubota's theta.

(c) [LIT] The quartic Gauss sums at squarefree moduli *are* the Whittaker coefficients of the distinguished theta on the 4-fold cover of $\mathrm{GL}_3(\mathbb{Q}(i))$, equivalently the ν-th Fourier–Whittaker coefficient $\psi^{(4)}(s,\nu) = \sum_c \tilde{g}_4(\nu,c)\mathrm{N}(c)^{-s}$ of Kubota's Eisenstein series on the 4-fold cover of GL_2 [10, 1, 2]; it has a meromorphic continuation and the functional equation of Diaconu [3], as used in [2, (4.28)] (with the normalised $\tilde{g}_4$ the pole is at $s = 3/4$).

(d) [DERIVED, from (c)] A rigorous transformation of (1) therefore exists, but with the wrong conductor. Since $\tilde{g}_4(h^3,c) = (h/c)_4\,\tilde{g}_4(1,c)$ for $(h,c)=1$ and $\tilde{g}_4(1,c)$ vanishes off squarefree c, (1) is a twisted $\psi^{(4)}$-type sum with index h^3, of conductor $\mathrm{N}(h)^3$; its functional equation carries the factor $\mathrm{N}(\nu)^{1-s}$ and sends column length X to dual length $\mathrm{N}(h)^3/X$ (cf. [2, §1.2], where an index a^{n-2} is said to "behave like a GL_{n-2}-object in a"; here the index is $h^{n-1} = h^3$), against $\mathrm{N}(h)^2/X$ for the cubic theta. The exponent 3 comes from the sign pairing: Poisson summation produces the index-h sums $\tilde{g}_1(h,n) = \overline{\chi_n(h)}\tilde{g}_1(n)$, and the absorption (iv) replaces $\tilde{g}_1$ by $\tilde{g}_{-1}$, the inverse character, which turns the pairing into the same-sign one, index h^3; in the sextic case absorption replaces γ_{-1} by γ_2, the square, and the sextic twist splits as $\overline{\chi_n^2(h)}\,(h/n)_2$, index h for the cubic part. At the scales of the argument, $X \asymp D$, $\mathrm{N}h \asymp D^{1-\vartheta}$, the dual length is $D^{2-3\vartheta} \gg X$: the transformation lengthens the sum. In the cubic case the dual length $D^{1-2\vartheta}$ is shorter than X by $D^{2\vartheta}$, which is what the quadratic large sieve converts into the target D^2. Under [2, (4.28)] the dual index is again h^3 (up to divisors), so the dual twist is the quartic $(h/c')_4$; the quadratic large sieve would not apply, and one would be left with the higher-order large sieve of Blomer–Goldmakher–Louvel with its $(ML)^{2/3}$ term. Thus both the conductor and the twist are worse than in the cubic case; the poles are not the issue (the residue at $s = 3/4$ is the theta coefficient at h^3, which vanishes for squarefree $h \neq 1$ by Suzuki [16, Thm. 4], see [2, (4.38)]).

(e) [LIT] A Voronoi formula for sums of quartic Gauss sums with level structure does exist [2, Lemma 9.2], but it is the one obtained from (4.28) and therefore carries the conductor of (d); no transformation with a GL_2-type conductor $\mathrm{N}(h)^2$ is known. Friedberg–Ginzburg's descent programme for the quartic GL_2 theta is conjectural [7].

The missing input, in one sentence. [JUDG] A transformation formula for (1) whose dual side has length $\approx \mathrm{N}(h)^2/X$ with explicit coefficients and a quadratic twist in h, i.e. an automorphic form on a cover of $\mathrm{GL}_2(\mathbb{Q}(i))$ whose Fourier coefficients at squarefree indices are the quartic Gauss sums $\tilde{g}_1(n)$; no such object is known: those Gauss sums are the Whittaker coefficients of the distinguished theta on the 4-fold cover of GL_3, with conductor $\mathrm{N}(h)^3$ under twisting, while the theta on the 4-fold cover of GL_2 is not distinguished and its coefficients at primes are unknown and conjecturally square roots of Gauss sums. Nothing in [2, 7] suggests this is within reach; the related quartic-bias conjecture for primes is described in [2] as "well out of reach" even under GRH (their unconditional result for primes is $X^{5/6+\varepsilon}$).

Remark 3 (quadratic over $\mathbb{Q}$)**.** *[JUDG] The order-2 family over $\mathbb{Q}$ would give $1-1/4 = 3/4$ at the first stage, but Poisson summation in u for $\sum_n \mu(n)(u/n)$ produces only explicit quadratic Gauss sums, so the dual of the Möbius-twisted quadratic family is again a Möbius-twisted quadratic family (nothing automorphic appears), and its amplifier rows $u = p^2$ are square moduli, on which the symbol is principal and which Theorem 1 of [8] excludes (it is stated for odd squarefree moduli, and must be). There is no analogue of Step 4 at all.*

Remark 4 (fields are not the lever)**.** *[LIT] Liu [12] obtains a uniform zero-free half-plane for finite-order Hecke L-functions over every number field, including $\mathbb{Q}(i)$ (abelian over $\mathbb{Q}$, so within his descent hypothesis), by running the 7/8-type cubic-theta argument over $K(\zeta_{12})$ and descending by class field theory; as far as its abstract shows, no quartic structure is used. The order of the residue symbol, not the field, sets the $1/6$.*

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